experiences moderate temperature and is therefore called as '**Temperate zone**'.

Polar Zone (Frigid Zone – 66 ½ °N to 90°N and 66 ½ °S to 90°S)

The region between North pole and Arctic Circle in the northern hemisphere and South pole and Antarctic Circle in the southern hemisphere is called 'Polar Zone'. This region always receives more oblique rays of the sun and so the temperature is very low. It is the coldest zone. This region experiences 24 hours of day and night during peak summer and winter respectively.



Annual Temperature : The average annual temperature of a region for a year.

Mean Annual Temperature: The average of 30 years of annual temperature of the region.

Range of Temperature: Difference between highest and lowest temperature of a place.

Annual Range of Temperature: The difference between highest and lowest temperature of a place in a year.

Diurnal range of Temperature: The difference between highest temperature and lowest temperature of a place in a day.

Green House Effect:

As seen in the heat budget, the longer wavelengths are absorbed by greenhouse gases in the atmosphere, increases the temperature of the atmosphere. These greenhouse gases act like a green house and retains some of the heat energy would otherwise be lost to space. The retaining of heat energy by the atmosphere is called the 'greenhouse effect'.

Global Warming

Global warming is observed in a centuryscale. The temperature increase over the years has been due to the greenhouse gas concentration such as carbon dioxide (CO_2), water vapour, methane and ozone. Greenhouse gases are those gases that contribute to the greenhouse effect. The largest contributing source of greenhouse gas is the burning of fossil fuels leading to the emission of carbon dioxide from industries, automobiles and domestic.

6.3.6 Urban Heat Island (UHI)

An urban heat island is an urban area or metropolitan area that is significantly warmer than its surrounding rural area due to high concentration of high rise concrete buildings, metal roads, sparse vegetation cover and less exposure of soil. These factors cause urban regions to become warmer than their rural surroundings, forming an "island" of higher temperatures (Figure. 6.9).

Ways to reduce the impact of urban heat island:

1. Increase shade around your home: Planting trees and other vegetation, provides shade and cooling effect through evapotranspiration and it lowers the surface and air temperature.

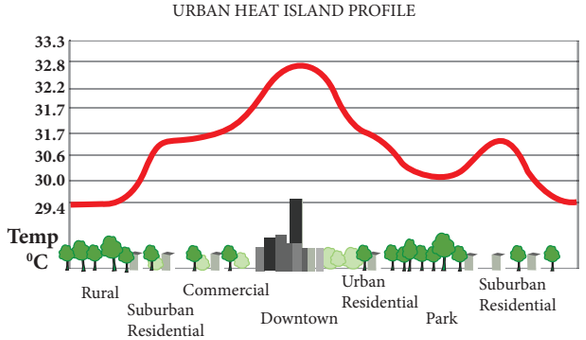


Figure 6.9 Urban Heat Island

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